ISOMETRIC

Transportation Emissions Accounting

Cross-pathway Accounting Module

v1.1

This module describes how transportation-related emissions must be calculated so that they can be subtracted in the net CO_2e removal calculation.

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1.0

Introduction

This module describes how transportation emissions should be considered for Carbon Dioxide Removal (CDR) projects, as part of project greenhouse gas (GHG) accounting. This module applies to all transportation modes, including transportation by trucks, rail, ships, pipeline and aircraft. Furthermore, this module applies to all CDR pathways, ensuring a consistently rigorous standard in how transportation emissions are quantified and reported between different CDR projects and approaches.

2.0

System Boundaries

The emissions associated with the transportation of goods must be calculated when any mode of transportation is used to move goods between sites as part of project operations. For a CDR project, the transportation of goods includes, but is not limited to, the transportation of CO_2, feedstock, consumables and wastes. The upfront transportation of materials associated with project establishment, for example to transport construction materials to site to build project facilities, is covered under the Embodied Emissions module.

At minimum, GHG emissions from the direct combustion of fuels in vehicles, as well as the upstream emissions associated with the production and distribution of the fuel and/or electricity must be accounted for.

A full cradle-to-grave GHG assessment is required for vehicles and infrastructure produced, constructed, and utilized for the CO_2 removal project. Where vehicles and/or infrastructure is not utilized explicitly for the CO_2 removal project, a proportional approach to emissions accounting can be taken.

3.0

Calculation of Transportation Emissions

3.1

Calculation Approach

Emissions are calculated as the sum of all transportation journeys, j, as part of a removal R:

$$CO_2e_{Transportation,R} = \sum_j CO_2e_{Transportation,j}$$

(Equation 1)

Where

$CO_2e_{Transportation,R}$ = the total quantity of GHG emissions associated with transportation of products for a removal R, in tonnes of CO_2e

$CO_2e_{Transportation,\,j}$ = the total quantity of GHG emissions associated with the transportation journey of a product, j, as part of a removal R, in tonnes of CO_2e , see below for calculations of this term

Equation (1) and Section 3 can also be followed for a batch n, or for a reporting period RP.

For each leg, $CO_2e_{Transportation,\,j}$ must be calculated by using one of the two approaches:

(1) Energy Usage Method: uses direct measurement of fuel or energy usage and associated emissions factors (preferred), see Section 3.2, Equation (2)

(2) Distance-Based method: uses the distance traveled, transport mode, and load weight with associated emissions factors (acceptable if data unavailable for (1)), see Section 3.3., Equation (3)

The Energy Usage Method must always be prioritized. Where it is not possible to use the Energy Usage Method due to lack of data and this is appropriately evidenced, the Distance-Based Method may be used.

3.2
Energy Usage Method

Emissions are calculated as follows:

$$CO_2e_{Transportation,\,j} = F_j \times EF_{Fuel,\,Transportation,\,j}$$

(Equation 2)

Where

$CO_2e_{Transportation,\,j}$ = the total GHG emissions associated with the transportation journey of a product, j, in tonnes of CO_2e

F_j = the quantity of fuel or electricity consumed during the transportation journey, j, from one location to another including other induced transportation, in appropriate units e.g. litres

$EF_{Fuel,\,Transportation,\,j}$ = the relevant fuel or electricity-based emission factor, in appropriate units e.g. kg CO_2e/litre

3.3
Distance-Based Method

Emissions are calculated as follows:

$$CO_2e_{Transportation,\,j} = D_j \times W_j \times EF_{Transportation,\,j}$$

(Equation 3)

Where

D_j = the distance traveled for the transportation journey, j, from one location to another, in appropriate units e.g. km

W_j = the mass of material transported as part of the transportation jounrey, j, from one location to another, in appropriate units e.g. tonnes

$EF_{Transportation,\ j}$ = the weight- and distance-based emission factor for transportation for a specific vehicle type, or infrastructure asset where available, provided in appropriate units e.g. kg CO_2e/tonne-km

3.4 Acceptable Emission Factors

For the Energy Usage Method, emission factors must:

- be for the specific type of fuel utilized in the vehicle (e.g. on-road diesel, biodiesel, gasoline, E-85, electricity)
- include all emissions associated with the fuel-cycle, such as direct combustion of fuel as well as indirect upstream emissions, including production and distribution of fuel or electricity

For the Distance-Based Method, emissions factors must:

- be selected for the specific vehicle type and age being utilized (e.g., Class 8 heavy-duty long-haul truck, Class 6 medium-duty truck, etc.)
- account for vehicle loading/capacity utilization (e.g., 50% capacity)
- include all emissions associated with the fuel-cycle such as direct combustion of fuel as well as indirect upstream emissions, including production and distribution of fuel or electricity

For both methods, emission factors must:

- be the most recently updated emission factor published
- account for total GHG emissions including, at a minimum, CO_2, CH_4, and N_2O emissions. Separate emission factors for each gas may be utilized and calculated emissions converted to CO_2e based on most recent IPCC 100-year global warming potentials

- be appropriately cited and described, and be from a Reputable Source in accordance with the Isometric Standard
- embodied life cycle emission factors must be used for vehicles and infrastructure produced, constructed, and utilized for the CO_2 removal project, and this must be calculated in additon to the Energy Usage or Distance-Based Methods. The requirements for emissions factors set out in the Embodied Emissions Module must be followed.

Recommendations for additional considerations for four common modes of transportation (road, rail, ship, and pipeline) are included in Appendix 1.

4.0

Book and claim units

Under this Module, the use of Book and Claim Units (BCUs) for low-carbon fuels to substitute for some, or all, of project fuel usage is permitted. BCUs are an instrument which Project Proponents can purchase to finance the use of low-carbon fuels by a third party in situations where the third party would otherwise have used conventional fuel. The net effect of the BCU purchase attempts yield the same outcome as if the Project Proponent had used low-carbon fuel within their own supply chain. BCUs can offer additional flexibility to Projects where constraints may limit availability of low-carbon fuels in the region of project operations. In the context of this Module “low-carbon fuels” refers to alternative transportation fuels with a lower carbon intensity than a conventional equivalent, for example biodiesel as a substitute for conventional diesel.

It should be noted that BCUs for low-carbon fuels may be used to substitute for both mobile and non-mobile fuel usage within the project system boundary.

4.1

Book and claim unit eligibility criteria

BCUs used to substitute for project fuel usage must meet all of the following eligibility criteria:

CRITERIA	DESCRIPTION	DOCUMENTATION REQUIRED
EC1	BCUs must be purchased and retired by the Project Proponent in a volume equivalent to the volume of fuel substitution claimed.	Provide proof of BCU purchase and retirement in a volume equivalent to the volume of fuel substitution claimed. Purchased BCUs must specify: • Quantity of low-carbon fuel usage represented by the BCU; • Feedstock and production process for the low-carbon fuel; • The location of low-carbon fuel production and/or usage; and • GHG emissions associated with low-carbon fuel production in an appropriate and usable set of units.
EC2	GHG emissions associated with low-carbon fuel production represented by BCUs must be verified in accordance with an existing regulatory or voluntary methodology.	Provide proof of verification of GHG emissions associated with low-carbon fuel production according to an existing methodology. Available methodologies vary by sector/application: • Aviation: The Roundtable on Sustainable Biomaterials (RSB) and the International Sustainability and Carbon Certification system (ISCC). Equivalent voluntary schemes exist (e.g. RSB Global and ISCC Plus), which may be more readily applicable in some settings. • Maritime: EU RED recognized methodologies. RSB and ISCC (or voluntary equivalents) are applicable for biofuels.

CRITERIA	DESCRIPTION	DOCUMENTATION REQUIRED
		• Other sectors: where suitable methodologies do not yet exist, a suitable alternative from another sector should be adopted for the purposes of verifying the GHG emissions associated with the fuel production supply chain. In such cases, a suitable approach will be agreed with Isometric at the point of project verification on a case-by-case basis.

EC3	Production and use of low-carbon fuels represented by BCUs must be demonstrably additional.	Provide either of the following:
		• Proof that the production facility associated with the low-carbon fuels represented in the BCUs entered into service no more than 36 months before initiation of The Project; or
		• Proof that revenue from BCU purchase by The Project is required for financial viability of production facility operations. For example, this could be satisfied by providing:
		  ◦ Production site financials demonstrating that BCU purchases are required to yield financial viability; or
		  ◦ A signed affidavit from the fuel producer stating that BCU purchases are required to yield financial viability.

EC4	The chain of ownership of all BCUs, from generation to retirement, is recorded in a suitable registry, or other suitable record keeping system.	Provide proof that BCU ownership, transference, and retirement is managed by a registry aligned with the principles established in the Book and Claim Communities "Best Practices for Book and Claim Systems in Heavy Transport" [1]. In circumstances where suitable registries have not yet been established, high quality record keeping is required to ensure traceability of ownership of the BCUs, and validation of verification of the fuel supply chain in accordance with the relevant methodology (see EC2). Project Proponents should refer to the Book and Claim Communities "Best Practices for Book and Claim Systems in Heavy Transport" (Principle 3) for record keeping requirements. Where BCUs are used off-registry, Project Proponents must provide a signed affidavit stating that the claimed BCUs will not be sold or otherwise used by any third party, or by the Project Proponent, in any other context.
EC5	A range of transportation decarbonization approaches should be considered for implementation by The Project.	Provide a statement that summarizes alternative decarbonisation options considered by the project in relation to project fuel usage. This should include a plan for future decarbonisation. Alternative decarbonisation opportunities can include, but are not limited to, the following: • Reducing transport distances; • Increasing efficiency by using less carbon intensive modes of transportation, for example rail instead of road; • Switching to electric, hybrid or hydrogen fueled transport alternatives; and • Purchasing low-carbon fuel for use in combustion engines.

4.2

Calculation of transportation emissions using book and claim units

When using BCUs to substitute for some, or all, of project fuel usage, the calculation approach described in the following subsections must be followed for the calculation of transportation emissions. Projects applying the energy usage method (as in Section 3.2) should follow the calculation approach described in Section 4.2.1 when using BCUs. Projects applying the distance-based method (as in Section 3.3) should follow the calculation approach described in Section 4.2.2 when using BCUs.

4.2.1

Calculation of transportation emissions using book and claim units with the energy usage method

When using BCUs to substitute for some, or all, of project fuel usage and calculating transportation emissions using the energy usage method, transportation emissions must be calculated as:

$$CO_2e_{Transportation,R} = \sum_j \left[EF_{Fuel,Transportation,j} \times \left(F_j - F_{BCU,j} \frac{ED_{BCU,j}}{ED_{Fuel,Transportation,j}} \right) + (EF_{BCU,j} \times F_{BCU}$$

(Equation 4)

Where:

- $F_{BCU,j}$ = the quantity of fuel represented in BCUs used for transportation journey j, in appropriate units e.g. liters.
- $EF_{BCU,j}$ = emissions factor of low-carbon fuel represented in BCUs used for transportation journey j, in appropriate units e.g. tonnes of CO_2e/liter.
- $ED_{BCU,j}$ = energy density of low-carbon fuel represented in BCUs used for transportation journey j, in appropriate units e.g. MJ/liter.
- $ED_{Fuel,Transportation,j}$ = energy density of fuel consumed during the transportation journey, j, in appropriate units e.g. MJ/liter.

When applying Equation 4, at maximum, an amount of BCUs may be used for each journey, j, such that:

$$\left(F_j - F_{BCU,j} \frac{ED_{BCU,j}}{ED_{Fuel,Transportation,j}} \right) \geq 0$$

(Equation 5)

4.2.2

Calculation of transportation emissions using book and claim units with the distance-based method

When using BCUs to substitute for some, or all, of project fuel usage and calculating transportation emissions using the distance-based method, the amount of fuel required for each transportation journey, j, must be calculated as:

$$F_j = D_j \times W_j \times \frac{EF_{Transportation,j}}{EF_{Fuel,Transportation,j}}$$

(Equation 6)

Where:

- $EF_{Transportation,j}$ = the weight- and distance-based emission factor for transportation for a specific vehicle type, or infrastructure asset where available, provided in appropriate units e.g. tonnes of CO_2e/tonne-km.
- $EF_{Fuel,Transportation,j}$ = emission factor of fuel assumed to be used for journey j, in appropriate units e.g. tonnes of CO_2e/liter. Assumption of fuel type used for journey j as a basis for this calculation must be based either (i) on information provided to the Project Proponent by the transport provider, or (ii) by prevailing fuel type used for the transport mode for journey j in the region of project operations, where the most granular or regionally-specific data possible is preferable.

Transportation emissions may then be calculated by following the approach established in Section 4.2.1 (see Equation 4), using the fuel usage for each transportation journey, F_j, calculated according to Equation 6.

This approach for the usage of BCUs is the closest proxy for using a low-carbon fuel as part of project operations. However, Project Proponents may elect to propose an alternative calculation method for the calculation of transportation emissions using BCUs, which must be specified in the public PDD and agreed with Isometric.

5.0

Measurements - $CO_2e_{Transportation, j}$

GHG emissions from transportation sources must be evaluated for all transportation completed between facilities, from the gate of one facility to the gate of the next facility. Primary measurements considered in calculation of emissions are:

- F_j (fuel/ electricity consumed)
- D_j (distance traveled)
- W_j (weight of vehicle load)

The fuel consumed, F_j , can be determined by one of the following methods:

- fuel/ electricity metering for vehicles, including data from calibrated on-board flow meters
- fuel/ electricity metering for pipeline transport
- fuel/ electricity usage data from fleet monitoring systems or software
- fuel/ electricity usage values provided by outputs of on-board vehicle diagnostic systems (OBD)
- fuel/ electricity efficiency (e.g. miles/gallon) data for transport vehicles used and distance travelled, to estimate fuel/ electricity use

The distance traveled, D_j, can be determined by one of the following methods:

- recording of vehicle odometer reading before and after completion of trip
- recording of travel distance by vehicle fleet management system
- online mapping of route traveled using common mapping platforms (e.g., Google Maps) and exact start and end trip locations
- other justifiable methods that account for actual route traveled for each shipment

Distance traveled and fuel usage must consider:

- full round trip distance of vehicle traveled when vehicle returns to origination site unloaded, or if next destination is unknown; or
- full distance of one-way trip plus distance of trip to the next destination

Evidence must be provided showing the distance of every trip to the next destination if the second option is used. When no onwards journey information is available, the full round trip must be assumed in calculations.

Calculations shall be completed separately for each leg of the trip associated with a removal, with total emissions calculated by the sum of emissions from each leg.

Weight of the material being shipped, W_j, should be determined as loaded gross vehicle weight measured at facility gate of departure minus vehicle gross weight upon facility entry, as determined by calibrated a weigh scale.

6.0

Required Records & Documentation - $CO_2e_{Transportation, j}$

Required records for each transportation leg calculated using the Energy Usage Method include:

- documentation of fuel consumed
- citation and description of emission factors used

Required records for each transportation leg calculated using the Distance-Based Method include:

- weigh scale tickets or similar documentation at each location to document load weight transported
- weigh scale calibration record
- bill of lading or similar transportation documentation indicating load type/contents, quantity, and pickup and delivery location
- documentation of vehicle destination after drop off, to account for determination of inclusion of return trip
- documentation of vehicle type and class used, including, whenever possible, vehicle (i.e., truck) class and model year

Any use of book and claim mechanisms to reduce reported transportation emissions should be transparently disclosed. Any meters used must be calibrated for the fuel being used both initially and at regular intervals in accordance with manufacturer specifications.

7.0

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8.0

Definitions and Acronyms

Cradle-to-Grave
Considering impacts at each stage of a product's life cycle, from the time natural resources are extracted from the ground and processed through each subsequent stage of manufacturing, transportation, product use, and ultimately, disposal.

Emission Factor
An estimate of the emissions intensity per unit of an activity.

GHG Assessment
The process by which all emissions associated with a Project's Removal or Reduction process, including leakages, are accounted for.

Global Warming Potential
A measure of how much energy the emissions of 1 tonne of a GHG will absorb over a given period of time, relative to the emissions of 1 ton of CO_2.

Greenhouse Gas (GHG)
Those gaseous constituents of the atmosphere, both natural and anthropogenic (human-caused), that absorb and emit radiation at specific wavelengths within the spectrum of terrestrial radiation emitted by the Earth's surface, by the atmosphere itself, and by clouds. This property causes the greenhouse effect, whereby heat is trapped in Earth's atmosphere (CDR Primer, 2022).

Module
Independent components of Isometric Certified Protocols which are transferable between and applicable to different Protocols.

Pathway
A collection of Removal or Reduction processes that have mechanisms in common.

Project
An activity or process or group of activities or processes that alter the condition of a Baseline and leads to Removals or Reductions.

Reputable Source
A source that would be widely considered trustworthy based on the process undertaken (e.g., peer review) or origin of the information (e.g., government body).

9.0

Relevant Works

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10.0

Appendix 1

Additional guidelines for choosing emission factors for different types of transportation modes.

Road:

- Resolution: The payload of the truck should be considered. Different emissions factors take payload into consideration differently. In general, higher payload utilizations translate to lower emissions on a per-metric-tonne-kilometer basis despite the increased fuel burn from transporting more mass.
- Timing: Due to relatively rapid changes in road transportation emissions regulation and technologies, utilized emissions factors should not be more than three years old. The most recent emissions factors available must be used.

Rail:

- Resolution: An emissions factor considering the source of the fuel should be used, with minimum resolution covering the use of diesel or electricity.
- Timing: Due to slower innovation in the locomotive space, utilized emissions factors should not be more than seven years old.

Ship:

- Resolution: Emissions factors matching the general type of ship (bulk carrier, tanker, barge, and cargo/container ship) and the general type of fuel (HFO, MDO, MGO, LNG, biodiesel, ammonia, hydrogen, or methanol) should be used.
- Timing: Utilized emissions factors should not be more than five years old.

Pipeline:

- Resolution: Most pipeline emissions factors are for natural gas and petroleum products and should be assigned accordingly. For transport of other products, direct energy needs for pipeline transportation should be calculated and multiplied by corresponding fuel-cycle emissions factors to estimate transportation impacts. If this is not possible, a petroleum product pipeline transportation emissions factor should be used. The natural gas factor is inflated due to consideration of methane leakage, which would not apply to most other products.
- Timing: Due to slower innovation in the pipeline space, utilized emissions factors should not be more than seven years old.

Aircraft:

- Resolution: Emissions factors for transportation via aviation should consider non-CO_2 impacts generated during combustion. Such impacts can arise from more than solely non-CO_2 GHGs due to the complex dynamics occurring during high-altitude combustion. The full impact can be determined using an appropriate radiative forcing multiplier. The UK Department for Energy Security and Net Zero recommends a multiplier of 1.7.[2]
- Timing: Utilized emissions factors should not be more than five years old.

1. https://bookandclaimcommunity.org/wp-content/uploads/2024/08/June-2024-BCC-Principles-and-Best-Practices-v1.1-July25.pdf ↩
2. https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2023 ↩

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Changelog

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CERTIFIED

Initial Release.

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Key requirement changes include the ability to use Book and Claim Units (BCUs) to substitute for some, or all, of project fuel usage.

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